NAGA SAI KRISHNA NELLURI

404-324-1029 nellurinagasaikrishna@gmail.com nelluri-naga-sai-krishna

Education

Kennesaw State University **January 2024 – May 2025**
Master of Science in Computer Science *CGPA: 3.80/4.00*
Relevant Coursework: *Network Security, Statistical Methods, Database Management Systems, Distributed Systems, Cloud Computing Architecture, Machine learning, Foundations of Software Engineering, Data Structures.*

Experience

Indian Servers - Software Development Company **May 2024 – August 2024**
Trainee-Machine Learning Intern *Vijayawada, India*

- Built and trained machine learning models on real-world datasets
- Applied techniques like feature engineering, model evaluation, and hyperparameter tuning.
- Collaborated on a mini-project focused on sentiment analysis.

Projects

Twin-AI: Personalized Conversational Agent (RAG Pipeline) | *github.com/krishna-53/Twin-AI* **Jan 2025 - April 2025**

- Developed an AI-powered personal assistant that ingests user audio, transcribes it using Faster-Whisper, and builds a searchable vector database.
- Engineered a RAG (Retrieval-Augmented Generation) pipeline using HuggingFace embeddings and ChromaDB to enable personalized, context-aware Q&A.
- Achieved ~40% improvement in retrieval relevance using overlapping text chunking and sentence-transformer embeddings.
- Deployed with FastAPI backend and Streamlit frontend, enabling seamless voice-to-query interaction.

Smart Grocery Inventory Assistant | *https://github.com/krishna-53/Smart-Grocery* **August 2024 - October 2024**

- Developed an AI-powered grocery inventory assistant using LangChain and GPT-4, reducing manual query resolution time by ~70% through automated product lookup.
- Built a custom LLM chain to generate context-aware answers from structured product data, achieving 90%+ accuracy in user query interpretation during testing.
- Integrated Streamlit frontend with a scalable backend, supporting real-time interactions with inventory containing 1000+ product records.

Fusion-based Hate Speech Detection | https://github.com/krishna-53/DL-Fusion-HSDetection **December 2022**

- Implemented a deep learning fusion framework for hate speech detection on SemEval Task 5 Twitter dataset, combining CNN, SVM (BERT), and ANN (ELMo) models.
- Enhanced model accuracy by **15%+** over standalone models using mean and max fusion of predictions, leveraging Word2Vec, BERT, and ELMo embeddings.
- Developed modular, reusable Python code enabling seamless integration of multiple embeddings and classifiers for robust text classification.

Technical Skills

Languages: Python, C++, C, HTML, CSS, JavaScript, Microsoft SQL Server, MySQL, FastApi, Streamlit
Developer Tools: VS code, Excel, PyCharm, Jupyter Notebooks
Technologies/Frameworks: TensorFlow, Keras, Scikit-learn, XGBoost, MLflow, GitHub, AWS Sagemaker, Docker
Data Handling & Analysis: Pandas, Time Series Analysis (ARIMA, SARIMA), MySQL, MongoDB, VectorDB (FAISS, ChromaDB)
Mathematics & Statistics: Applied Math, Statistical Modeling

Publications

Breast Cancer Detection Using Pre-Processed Images

Published in AAIOT 2023 | https://github.com/krishna-53/publication **April 2023**

- Designed and implemented a breast cancer detection system using pre-processed mammographic images to enhance diagnostic precision.
- Applied advanced image enhancement techniques for noise reduction, background removal, and contrast normalization.
- Developed and compared CNN, KNN, and Decision Tree classifiers, achieving 95%, 90%, and 85% accuracy respectively.
- Employed ensemble learning with mean fusion to boost overall accuracy to 97%.
- Used the CBIS-DDSM dataset comprising over 10,000 labeled mammographic images.